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Towards operational atmospheric correction of airborne hyperspectral imaging spectroscopy: algorithm evaluation, key parameter analysis, and machine learning emulators

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Abstract

Atmospheric correction of airborne hyperspectral imaging spectroscopy (AHIS) to obtain high-quality surface reflectance is the prerequisite for remote sensing applications. Over the last decades, different atmospheric correction methods have been developed based on radiative transfer models (RTMs), however, the relative performances of different algorithms are unclear. Automated operational atmospheric correction methods to process large-volume AHIS data in a high-accurate and high-throughput manner are still lacking. Therefore, this study proposed an operational atmospheric correction pipeline for deriving surface reflectance from AHIS data. To ensure the accuracy and efficiency of the pipeline, we focused on three specific aspects: (1) selecting a suitable RTM for the

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development of atmospheric lookup tables (LUTs) by comparing the commercial MODerate resolution atmospheric TRANsmission (MODTRAN) and open-sourced Library for Radiative TRANsfer (LibRadTRAN) models, where the widely-used software, Atmospheric/Topographic Correction for Airborne Imagery (ATCOR), was used as benchmarks; (2) identifying key atmospheric correction parameters and determining suitable sources for parameter retrievals including AHIS, Moderate Resolution Imaging Spectroradiometer (MODIS), and AErosol RObotic NETwork (AERONET); and (3) testing the performance of using machine learning emulators to speed up the RTM-based atmospheric correction. Results indicate that (1) atmospheric correction based on MODTRAN LUTs can produce surface reflectance accurately with mean absolute errors <0.05 and cosine similarities >0.98 compared to field measurements, which is comparable to the software ATCOR and slightly outperforms the LibRadTRAN LUTs; (2) sobol global sensitivity analysis demonstrates that in the atmospheric correction, visibility and water vapor are two key parameters that can be accurately derived from AHIS in contrast to MODIS or AERONET data, and (3) Random Forest emulators can produce accurate estimations of surface reflectance with mean absolute errors <0.03 and cosine similarities >0.98 for higher processing efficiency and determine a suitable set of wavelengths for retrieving atmospheric visibility and water vapor. The proposed atmospheric correction pipeline also improved the four-stream radiative transfer theory for airborne applications by considering adjacent effects from airborne surrounding pixels and can also be applied for atmospheric correction of hyperspectral data from spaceborne missions.

Keywords:

Hyperspectral; Atmospheric correction; Surface reflectance; Radiative transfer modeling; Machine learning

1. Introduction

Airborne hyperspectral imaging spectroscopy (AHIS) in the optical domain of 400–2500 nm with a spatial resolution of < 1 m provides unprecedented opportunities to monitor vegetation traits and

soil properties for both natural and agricultural ecosystems with high spatial and spectral resolutions (Asner et al., 2015; Wang et al., 2019; Zarco-tejada et al., 2016). However, the signal detected by airborne remote sensing sensors contains information of both the Earth's surface and atmosphere (Rani et al., 2017). To map the properties of the Earth's surface accurately, the applications of AHIS typically begin by converting the raw digital number (DN) to calibrated at-sensor radiance through radiometric calibration, and further to surface reflectance via atmospheric correction (Dao et al., 2019; Thompson et al., 2018). Atmospheric correction of AHIS is essential to remove atmospheric absorption and scattering effects from imaging spectrometer data before studying surface properties (Gao et al., 2009; Thompson et al., 2019).

An operational atmospheric correction pipeline is highly needed for processing large-volume AHIS data in a high-accurate and high-throughput manner. Atmospheric correction algorithms of AHIS can be divided into three categories: (1) empirical methods, such as the Internal Average Reflectance (IAR) approach (Kruse, 1988), "cloud shadow" method (Lee et al., 2005), and Empirical Line Calibration (ELC) approach (Karpouzli and Malthus, 2003); (2) inversion of radiative transfer modeling (RTM) approaches, such as the Fast Line-of-sight Atmospheric Analysis of Spectral Hypercubes (FLAASH) (Cooley et al., 2002), Atmosphere CORrection Now (ACORN) (Kruse, 2004), and Atmospheric/Topographic Correction (ATCOR) (Richter and Schläpfer, 2016); and (3) machine learning approaches, such as the Gaussian processes regression (Servera et al., 2021) and time-dependent deep neural network for atmospheric correction (Sun et al., 2021). The inversion of RTM methods or RTM-based machine learning approaches, which can provide reasonable surface reflectance without requirements of intensive concurrent field measurements and training or calibration datasets. These methods are more suitable for large-volume data processing of AHIS because they can be operationally implemented for atmospheric correction under various weather conditions. Therefore, developing operational RTM-based atmospheric correction pipeline, which is based on inversion or machine learning surrogate modeling, is desired for reliable and automated data processing of AHIS.

The RTM inversion approach for atmospheric correction is affected by multiple factors including the selection of RTMs, the inversion approaches, and the RTM parameters. The MODerate spectral resolution atmospheric TRANsmittance (MODTRAN) (Berk et al., 1987) and Library for

Radiative TRANsfer (LibRadTRAN) (Emde et al., 2016) are widely used two commercial and open-source atmospheric RTMs, respectively. Studies have conducted and compared atmospheric correction for processing AHIS data based on the two models (Giardino et al., 2015; Katkovsky et al., 2018; Kereszturi et al., 2018). For example, Katkovsky et al. (2018) found that the FLAASH algorithm based on MODTRAN4 often gives negative values of spectral reflectance, while the Satellite Hypercube Atmosheric Rapid Correction (SHARC) based on LibRadTRAN represents a better version of atmospheric correction. Regarding commercial RTM-based atmospheric correction software, ATCOR based on MODTRAN has been proven accurate to produce surface reflectance for various hyperspectral sensors (Giardino et al., 2015; Kereszturi et al., 2018). In the inversion of RTM, the four-steam approach standardizes the simulation outputs from RTMs to solutions of atmospheric transfer functions (Verhoef and Bach, 2007). This approach has shown high accuracies for atmospheric correction to obtain high-quality optical hyperspectral reflectance, solar-induced fluorescence, and thermal infrared data (Bayat et al., 2020; Verhoef et al., 2018; Yang et al., 2020b). For the relatively homogenous land surface (soil and crops) in the U.S. Midwest, the four-stream theory is a reasonable approximation for atmospheric correction (Verhoef and Bach, 2012, 2007, 2003) for AHIS images. Therefore, we have selected the four-stream theory for atmospheric correction of AHIS in the U.S. Midwest based on commercial MODTRAN and open-sourced LibRadTRAN for RTM simulation, where the widely used ATCOR is used as benchmarks.

The input parameters for RTMs strongly influence the accuracy and efficiency of RTM-based atmospheric correction. Parameters related to flight paths (e.g. sensor altitude and ground elevation), atmospheric conditions (e.g. visibility, water vapor, ozone, and carbon dioxide), and imaging geometry (e.g. relative azimuth angle, Sun zenith angle, and view zenith angle) (Berk and Hawes, 2017) are required for atmospheric correction of AHIS. Evaluating the sensitivity and importance of each parameter in RTMs can help identify the key parameters and assess their influence on the final prediction of surface reflectance (Prikaziuk and van der Tol, 2019). RTM lookup tables (LUTs) simulated by these required parameters can speed up RTM simulations (Richter and Schläpfer, 2016) for atmospheric correction. To use the LUTs accurately and efficiently, these input parameters should be selected carefully and obtained reliably. For example, the visibility and water vapor can be derived from multiple

sources such as AHIS radiance data (Thompson et al., 2019), satellite products (e.g. MODIS, (King et al., 2003)), and AERONET observations (Cheng et al., 2012). Comparisons of parameter retrievals from these sources are beneficial for optimal RTM parameterization.

Even with precalculated LUTs, RTM-based atmospheric correction algorithms are relatively time-consuming (Duffy et al., 2019); alternatively, advanced machine learning models have the potential to speed up the atmospheric correction pipeline. Machine learning has been widely applied in the remote sensing community for terrestrial, oceanic, and atmospheric applications (Lary et al., 2016), which offers the potential to develop efficient, complex, and nonlinear approximations of physical models (Duffy et al., 2019). Since traditional RTM-based atmospheric correction is time-consuming, it is of great interest to build machine learning emulators to atmospherically correct AHIS data for higher efficiency (Duffy et al., 2019; Shamshiri et al., 2020).

Motivated by the need for accurate and efficient data processing of large-volume AHIS data, our objective was to develop an automated and operational pipeline for the atmospheric correction of AHIS data. Specifically, for the accuracy and efficiency of the pipeline, this study focused on three major objectives. Objective 1: selecting a suitable RTM for atmospheric correction by evaluating the commercial MODTRAN and open-sourced LibRadTRAN. The well-performing commercial software, ATCOR, was used as the benchmark. This evaluation can provide insights into RTM selections for atmospheric correction of AHIS. Objective 2: determining the key parameters for RTM simulation based on sensitivity analysis and assessing the impacts of parameter retrievals from multiple data sources. This analysis and assessment can contribute to the development of RTM LUTs and the selection of reliable sources for parameter retrievals. Objective 3: quantifying the potential of machine learning techniques for atmospheric correction of AHIS benchmarked with the results from RTM-based algorithms. Since the RTM-based atmospheric correction for large-volume AHIS is time-consuming, the machine learning emulators may speed up data processing and improve efficiency.

2. Study area and data

2.1. Study area

The study area is located in Champaign County in central Illinois (40°0′ N–40°4′ N, 88°10′ W–88°18′ W, see Fig. 1), which is one of the major U.S. agricultural states. Champaign County is in a humid continental climate with a mean annual temperature of 10 °C and precipitation of 1008 mm, and an average growing season of 185 days. Vegetation and soil are the primary land cover types in the flight paths for this study (Fig. 1), with agriculture (mostly corn and soybean) accounting for approximately 96.5% of the area.

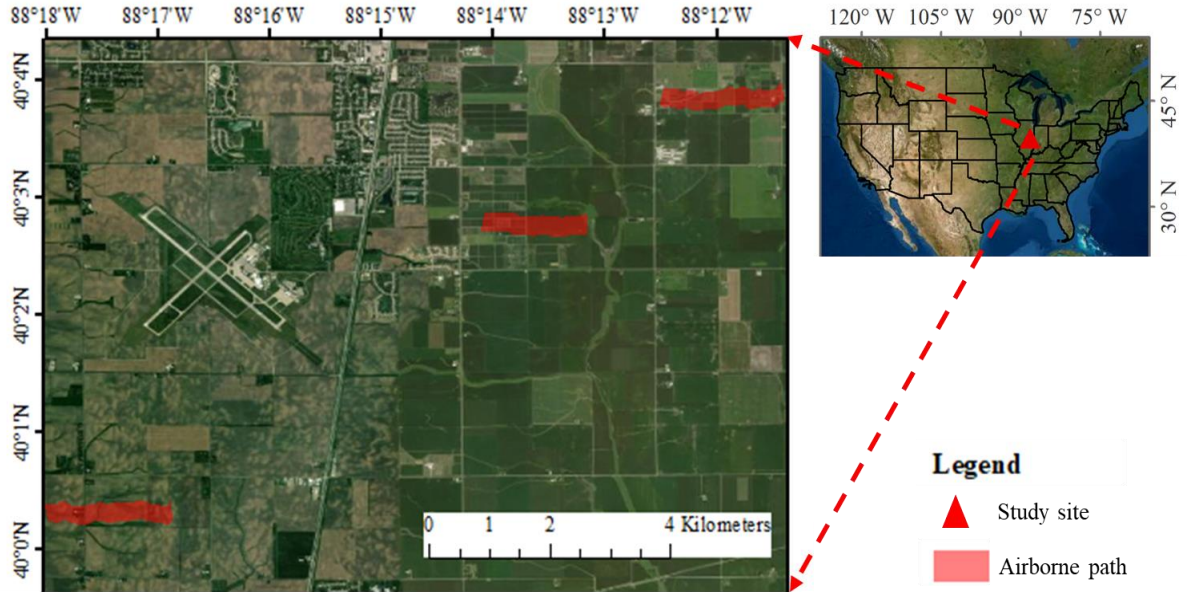


Fig. 1. Overview of the study site (red triangle) and collected airborne hyperspectral imaging spectroscopy (AHIS) flight paths (red polygons) on June 11th and 18th, 2020 in Champaign County, Illinois.

2.2. Ground observations

Three tarps representing high, medium, and low reflective properties (spanning 10-80% reflectance) were used to collect field hyperspectral measurements (400–2500 nm with an interval of 1 nm). The tarps were placed in the airborne paths during the flight campaigns on June 11th and 18th, 2020 under clear sky conditions. For each tarp, the spectra were measured six times within minutes to reduce the measurement uncertainties using a Spectral Evolution PSR + spectroradiometer (Spectral Evolution

Inc., Lawrence, Massachusetts, USA) and a 99% white panel as reference reflectance (Labsphere, North Sutton, NH).

2.3. Airborne data

The raw AHIS data were collected by a HySpex VNIR-1800 and SWIR-384 system (HySpex NEO, Skedsmokorset, Norway). The combined spectral coverage is 400–2500 nm with an interval < 5.5 nm (The main specifications of the two sensors can be found in Table S1). The imaging system was coupled with a data acquisition unit, a differential Global Navigation Satellite System (GNSS) and Inertial Measurement Unit (IMU) navigation system (Trimble, Georgia, USA), and a spring-dampened mounting platform. The IMU and GPS information was processed using POSPac (Applanix, Richmond Hill, ON, Canada) and HySpex NAV (Norsk Elektro Optikk, Skedsmokorset, Norway) after collecting raw AHIS data. The axes of VNIR and SWIR sensors were co-aligned to remove co-registration errors, and both sensors were triggered with the same trigger scaled down by a factor 2 for the SWIR sensor, resulting in a VNIR image with twice the resolution. The navigation system was boresight calibrated, making it simple for direct geo-referencing without ground control points. The AHIS data were collected on June 11th and 18th, 2020 under clear sky conditions (Table 1). The flight height was about 900 m over the ground, leading to a spatial resolution of approximately 0.4 m and 0.8 m for VNIR and SWIR sensors, respectively. Boresight calibration was done by applying boresight parameters, which were obtained using ground control points. Smile effects were corrected based on spectral absorption features (Richter and Schläpfer, 2016). Radiometric calibration was conducted by applying manufacturer laboratory-based calibration coefficients to convert the raw DN of AHIS to at-sensor radiance, and the geometric correction was done through ray-tracing-based geographic LUTs (Ferreira et al., 2014; Liu et al., 2019). After these steps, we can obtain geometrically corrected at-sensor AHIS radiance.

Table 1. Airborne hyperspectral flight campaigns on June 11th and 18th, 2020 under clear sky conditions. The time is the local central standard time (CST).

Date	Acquisition time (HH: MM: SS)	Height above the ground (m)	Air temperature[1] (°C)	Wind speed[1] (m/s)	Humidity[1] (%)	AOD[2]
June 11th	14:19:00	860	28.3	6.3	25	0.065
June 18th	10:51:29	925	28.3	1.3	39	0.243

[1]Air temperature, wind speed, humidity are acquired from the Weather Underground (https://www.wunderground.com/).

[2]AOD (Aerosol optical depth) is obtained from the AERONET station at Bondville, IL: (40.053N, 88.372W; https://aeronet.gsfc.nasa.gov/), about 12 km from the study sites.

3. Methodology

3.1. Development of an automatic and operational data processing pipeline of AHIS

The data processing pipeline for atmospheric correction of AHIS to generate surface reflectance products is illustrated in Fig. 2. To convert the georectified at-sensor radiance to surface reflectance, the atmospheric correction of AHIS in our pipeline has two options: (1) based on traditional RTM LUTs and (2) based on machine learning emulators. In this step, key atmospheric parameters (e.g. visibility and water vapor) were retrieved and then the atmospheric correction was performed based on the four-stream theory with RTM LUTs or machine learning.

To understand the uncertainty of parameters to atmospheric correction performance, we conducted a global sensitivity analysis of model parameters to identify key parameters and then compared the multiple sources of atmospheric parameter values. Specifically, visibility and water vapor can be derived from multiple sources including the AHIS data, satellite products, or ground observations. From AHIS, visibility can be estimated using dense dark vegetation (DDV) pixels and water vapor can be derived using the atmospherically precorrected differential absorption (APDA) method (Richter and Schläpfer, 2016). The MODIS satellite products can provide an estimation of atmospheric visibility and water vapor. The mean values of nine valid pixels nearest to the concurrent AHIS scene are used. The

AERONET (nearest site Bondville: 40.053°N, 88.372°W) site provides ground observations of atmospheric visibility and water vapor.

To derive surface reflectance, four-stream theory with MODTRAN (version 5) and LibRadTRAN (version 2) simulations were both applied with the inputs of AHIS radiance, atmospheric parameters (e.g. visibility and water vapor), imaging geometry (viewing and zenith angles), and flight path (flight altitude and ground elevation). With atmospheric parameters, imaging geometry, and flight path, both MODTRAN and LibRadTRAN could provide simulations of albedos of 0, 50%, and 100%. Compared with AHIS radiance, surface reflectance could be estimated using the four-stream theory (more details refer to section 3.2.1). Since each AHIS scene is small or divided into small pieces (~5 km* 5 km), the mean values of atmospheric parameters, imaging geometry, and flight path were used for the whole scene for RTM simulation. For the efficiency of the pipeline, machine learning-based atmospheric correction is also tested. One submodel (#1) was developed to retrieve atmospheric parameters and another submodel (#2) was developed to derive surface reflectance after obtaining atmospheric parameters (more details refer to section 3.2.3). Similar to RTM-based atmospheric correction, the mean values of visibility and water vapor, flight path, and imaging geometry were used for the whole scene.

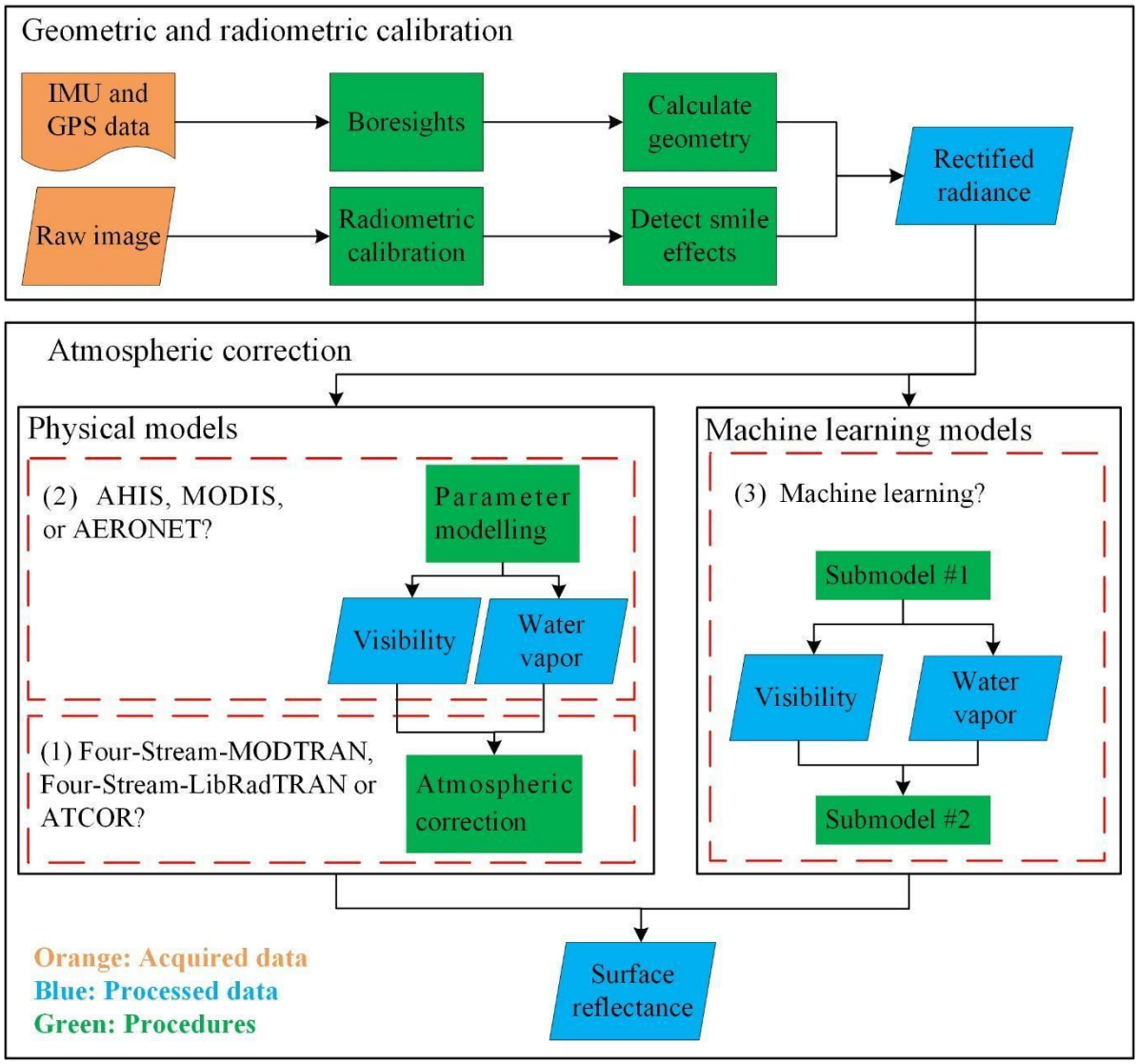


Fig. 2. Automatic and operational data processing pipeline of AHIS. The red rectangles with dash lines indicate three main objectives in the study: (1) to evaluate Four-Stream-MODTRAN, Four-Stream-LibRadTRAN, and ATCOR for atmospheric correction of AHIS; (2) to determine the sensitivity of input parameters and assess the impacts of visibility and water vapor from the AHIS, MODIS, and AERONET observations on the atmospheric correction; and (3) to quantify the efficiency of machine learning-based atmospheric correction, which includes two submodels for deriving visibility and water vapor, and surface reflectance, respectively.

3.2. RTM-based atmospheric correction

The four-steam radiative transfer theory with MODTRAN and LibRadTRAN LUTs was used for the atmospheric correction of AHIS. The atmospheric correction parameters (e.g. visibility and water

vapor) were firstly derived (section 3.2.3). With these input parameters, RTM simulations were obtained from pre-simulated RTM LUTs (section 3.2.2). With RTM simulations, the surface reflectance can be derived from at-sensor radiance using the four-stream theory (section 3.2.1).

3.2.1. Four-stream surface-atmosphere radiative transfer modeling of AHIS

The four-stream surface-atmosphere radiative transfer theory (Verhoef, 1985; Verhoef et al., 2018; Verhoef and Bach, 2012, 2007, 2003) is a powerful and flexible tool for the simulation of atmospheric effects and atmospheric correction of satellite imagery. However, observations from an airborne platform are more complicated than satellite observations, because the sensor is located inside the atmosphere, with one part above the plane (layer 1) and another part below it (layer 2). After modification of the four-stream theory, for a Lambertian surface, the signal received by an airborne sensor can be written as (see Appendix for details):

$$L_{at-sensor} = L_{path} + \frac{G_t r_t + G_b r_b}{1 - r_{bS}} \quad (1)$$

where L_{path} is the path contribution; G_t and G_b are the gain factors for target and background, respectively; S is the spherical albedo of the atmosphere; r_t and r_b are surface reflectance of target and background, respectively. Thus, the relationship between $L_{at-sensor}$ and target reflectance of high-resolution airborne hyperspectral data can be determined by Eq. (1).

To calculate parameters for atmospheric correction in Eq. (1), Verhoef and Bach (2012, 2007, 2003) developed the MODTRAN Interrogation Technique, which uses RTM simulations to estimate S, G_t, G_b, and L_{path} in Eq. (1). More details of applying the MODTRAN Interrogation Technique to AHIS data can refer to the Appendix. However, these outputs do not include r_b, which can be calculated based on spatial filtering or just the whole scene average (Fawcett et al., 2018). With the derived parameters from RTM and estimated r_b, we can obtain Eq. (2) with all parameters solved:

$$r_t = \frac{(L_{at-sensor} - L_{path})(1 - r_b S) - G_b r_b}{G_t} \quad (2)$$

For homogenous and flat land covers ($r_t \approx r_b = r$), Eq. (2) can be simplified as:

$$r = \frac{L_{at-sensor} - L_{path}}{G_t + G_b + S(L_{at-sensor} - L_{path})} \quad (3)$$

The RTM outputs provide the necessary parameters in Eq. (3). If the adjacent effects are ignored, only simulations of albedos of 0, 50%, and 100% are needed to solve the parameters in Eq.(3).

$$L_{path} = L_{00} \tag{4.1}$$

$$\mathrm{S} = \frac{L_{100}+L_{00}-2L_{50}}{L_{100}-L_{50}} \tag{4.2}$$

$$G_t + G_b = (L_{100} - L_{00})(1 - \mathrm{S}) \tag{4.3}$$

where L_{00}, L_{50}, L_{100} are the total reflected radiance from RTM simulations of albedos of 0, 50%, and 100%, respectively. With RTM simulations, atmospheric LUTs can be established for faster atmospheric correction based on the four-stream theory.

3.2.2. MODTRAN and LibRadTRAN LUTs

For the calculation of surface reflectance of AHIS, three RTM simulations of albedos of 0, 50%, and 100% are needed for the four-stream theory (Guanter et al., 2009; Haan and Kokke, 1996; Verhoef and Bach, 2003; Williams et al., 1992). MODTRAN and LibRadTRAN can conduct the simulations with necessary inputs such as flight height, imaging geometry, and atmospheric parameters (Berk and Hawes, 2017; Emde et al., 2016; Verhoef and Bach, 2012). To speed up the RTM simulations, predefined LUTs were utilized (Liang et al., 2006; Zhou et al., 2020a). With the RTM LUTs, simulations under various conditions can be calculated via interpolation. The input parameters and their values to build atmospheric LUTs using MODTRAN and LibRadTRAN are summarized in Table 2. The simulated total reflected radiance was written into the LUTs.

Table 2. Input parameters and values of atmospheric lookup tables (LUTs).

Parameters	Unit	Values
albedo	none	[0.0, 0.5, 1.0]
Sensor altitude	m	[800, 1300, 4000]
Ground elevation	m	[100, 500]
Visibility	km	[5, 10, 20, 40, 80, 120]
Water vapor	g/cm^2	[0.4, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0]

View zenith angle	degree	[0, 5, 10, 15, 20, 25, 30, 40]
Relative azimuth angle	degree	[0, 15, 30, 45, 60, 75, 90, 105, 120, 135, 150, 165, 180]
Sun zenith angle	degree	[0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70]

3.2.3. Retrieval of atmospheric correction parameters

Sensitivity analysis of atmospheric correction parameters is essential to build and utilize RTM LUTs. Two approaches for sensitivity analysis were applied: (1) a global approach based on Sobol's sensitivity analysis (Sobol, 1993) and (2) a local analysis by fixing one parameter and changing others. The parameter ranges for Sobol's sensitivity analysis are shown in Table 3. The comprehensive conditions are the general setting of ATCOR, and the specific conditions are determined from an AHIS observation. The comprehensive condition represents general conditions for AHIS collections, which cover the parameter ranges for most AHIS imaging situations. In these conditions, we considered all these parameters as unknown variables. However, the specific condition represents the conditions for a certain AHIS image. The sensor altitude, ground elevation, and imaging angles are measured or can be calculated, which were considered as known variables with certain measurement uncertainties. Visibility water vapor, ozone, and carbon dioxide were unknown variables in this condition. We used the Python package SALib (Herman and Usher, 2017) for Sobol's global sensitivity analysis. For the local sensitivity analysis, the value of the fixed parameters was obtained from ATCOR, and the other parameters were changed accordingly. The surface reflectance resulting from the varied combination of the parameters was used to determine the parameter impacts.

Table 3 Inputs for Sobol's sensitivity analysis under comprehensive and specific conditions. The comprehensive conditions refer to parameters that are assumed for general conditions, while specific conditions have some measured parameters with known uncertainties for a specific AHIS flight line.

Parameters	**Name**	**Unit**	**Comprehensive range[6]**	**Specific range[7]**
At-sensor radiance[1]	RAD	mW $cm^{-2}nm^{-1}sr^{-1}$	[0, Maximum]	Mean[1−1/SNR, 1+1/SNR]
Sensor altitude[2]	SAL	m	[200, 2000]	[1090.5−0.05, 1090.5+0.05]

Ground elevation[3]	ELE	m	[100, 500]	[216−30, 216+30]
Visibility	VIS	km	[0, 120]	[0, 120]
Water vapor	H_2O	g/cm^2	[0, 5.0]	[0, 5.0]
Carbon dioxide	CO_2	ppmv	[300, 500]	[300, 500]
Ozone	O_3	g/cm^2	[6.0*10^{-6}, 6.5*10^{-6}]	[6.0*10^{-6}, 6.5*10^{-6}]
View azimuth angle[4]	VAA	degree	[0, 360]	[252.1−0.1, 252.1+0.1]
View zenith angle[4]	VZA	degree	[0, 30]	[6.2−0.1, 6.2+0.1]
Sun azimuth angle[5]	SAA	degree	[0, 360]	[111.5−0.5, 111.5+0.5]
Sun zenith angle[5]	SZA	degree	[0, 70]	[32.0−0.5, 32.0+0.5]

[1]The uncertainty of at-sensor radiance is estimated by sensors' SNR (see Table S1) and the at-sensor radiance is obtained from all airborne paths in this study (see Fig. S1);

[2]Sensor altitude is measured with an uncertainty <0.05 m;

[3]Ground elevation is from global multi-resolution terrain elevation data 2010 at 7.5 arc-seconds spatial resolution with an uncertainty <30 m;

[4]View angles are measured with an uncertainty <0.01 degree;

[5]Sun angles are calculated with an uncertainty of <0.5 degree (Blanco-Muriel et al., 2001);

[6]Comprehensive ranges are adopted from atmospheric lookup tables, except the radiance is calculated;

[7]Specific ranges are from measured data with uncertainties unless the parameters are not measured during the flight plan (e.g. VIS, H_2O, CO_2, and O_3).

The sources to derive the atmospheric parameters are also important for atmospheric correction of AHIS. MODIS aerosol (MYD04 and MOD04) and water vapor (MYD05 and MOD05) products and AERONET data could provide direct aerosol and water vapor estimations. The observations from synchronous MODIS products and the nearest AERONET site (Bondville, IL, USA) on the same day were tested here. For the AHIS data, visibility was derived based on the DDV pixels and the known spectral correlation of surface reflectance between certain bands. The DDV pixels were determined with $0.01 < \rho_{2200} < 0.05$ and NDVI > 0.1; and the relationship of $\rho_{660} = 0.5\,\rho_{2200}$ and $\rho_{480} = 0.5\,\rho_{660} + 0.005$ was employed to estimate visibility automatically. Water vapor derived from the AHIS data was based on the APDA method (Richter and Schläpfer, 2016).

3.3. Machine learning-based atmospheric correction

The pipeline for atmospheric correction of AHIS based on machine learning includes two submodels: (1) retrieval of visibility and water vapor, and (2) estimation of surface reflectance after deriving visibility and water vapor. We implemented the Random Forest algorithm (Breiman, 2001) to build the two submodels. The input of submodel #1 was the total radiance for all the AHIS bands, and the output was visibility and water vapor; the input of submodel #2 was single-band AHIS radiance, visibility and water vapor, flight path, and imaging geometry, and the output was surface reflectance of the corresponding band. The visibility, water vapor, surface reflectance, and total reflected radiance from MODTRAN simulations were used as "truth data" to train and validate the two submodels. Over 60,000 samples were simulated with randomly selected inputs for training and validation using a ten-fold cross-validation strategy. For submodel #1, the inputs were simulated at-sensor radiance for all the AHIS bands, imaging geometries, and visibility/water vapor, and the output was water vapor/visibility. For actual applications, the visibility and water vapor were estimated iteratively until both visibility and water vapor were stable. For submodel #2, the inputs were at-sensor radiance for each band, visibility, water vapor, and imaging geometries, the output was surface reflectance for a Lambertian surface at the corresponding band. Submodel #2 was built for individual bands of AHIS to avoid band dependence.

3.4. Performance assessment

Model performance was evaluated using the coefficient of determination (R^2) and root mean square error (RMSE), as well as the mean absolute error (MAE, Eq. 5) for spectral values and cosine similarity (CS, Eq. 6) for spectral shapes:

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|\rho_T(i) - \rho_e(i)| \quad (5)$$

$$CS = cos^{-1}|\frac{\rho_T \cdot \rho_e}{||\rho_T||\,||\rho_e||}| \quad (6)$$

where ρ_T and ρ_e are the vectors of the 'truth' and estimated hyperspectral reflectance, respectively; N and i are the total bands and the i_{th} band of the hyperspectral reflectance, respectively. Both the ground

tarp measurements and results from ATCOR were used as references. The settings of ATCOR in this study can refer to Text S1.

4. Results

4.1. Performance of RTM-based atmospheric correction of AHIS

The reflectance values of tarp measurements (400–2500 nm with an interval of 1 nm) covered a large range from 0.1 to 0.8 (Fig. 3, black lines), which are sufficient to validate the performance of atmospheric correction over targets from low to high reflective properties. The similarity of measurements on June 11th and 18th, 2020 (first and second columns in Fig. 3) indicates that the tarps were stable and the small standard deviation values (grey regions in Fig. 3) demonstrate the reliability and accuracy of the field measurements. The tarp hyperspectral measurements have large noises in the atmospheric water absorption bands from 1800 nm to 2000 nm (black rectangles in Fig. 3), thus these bands were excluded in the analysis.

The length of airborne flight lines varies greatly due to differing flight plans and imaging conditions, which results in different area sizes to represent background reflectance for consideration of adjacency effects in AHIS data. The surface reflectance from the four-stream theory with background reflectance taking from the average of different sizes from 20×20 m^2 to the whole scene (approximately 5000×5000 m^2) are shown in Fig. S2. Overall, the accuracy is stable and just slightly affected by the size of the background area (see Fig. S2). However, the size corresponding to the best performance varies for different tarps and dates, indicating that taking the scene average as background reflectance may affect the results of atmospheric correction. This may be attributed to the fact that the length of AHIS data is much larger than the width (see Fig. 1), while the signal received by the AHIS sensors is affected by background targets from all directions. For relatively homogeneous regions such as the agricultural site in this study, the surface reflectance from the four-stream theory assuming uniform (Eq. (A9)) or non-uniform (Eq. (A13)) surfaces are similar (see Fig. S3). Thus, it is acceptable to use the solution of the four-stream theory for a uniform surface for atmospheric correction of AHIS data in relatively homogeneous agricultural regions. All three methods (Four-Stream-MODTRAN, Four-Stream-

LibRadTRAN, and ATCOR) can produce reasonable surface reflectance from raw AHIS data with small MAE (<0.06) and large CS (>0.99) from high (tarp 1) to medium (tarp 2) to low (tarp 3) reflective properties (Fig. 3). For the three approaches, the Four-Stream-MODTRAN has similar performance to ATCOR and slightly outperforms Four-Stream-LibRadTRAN.

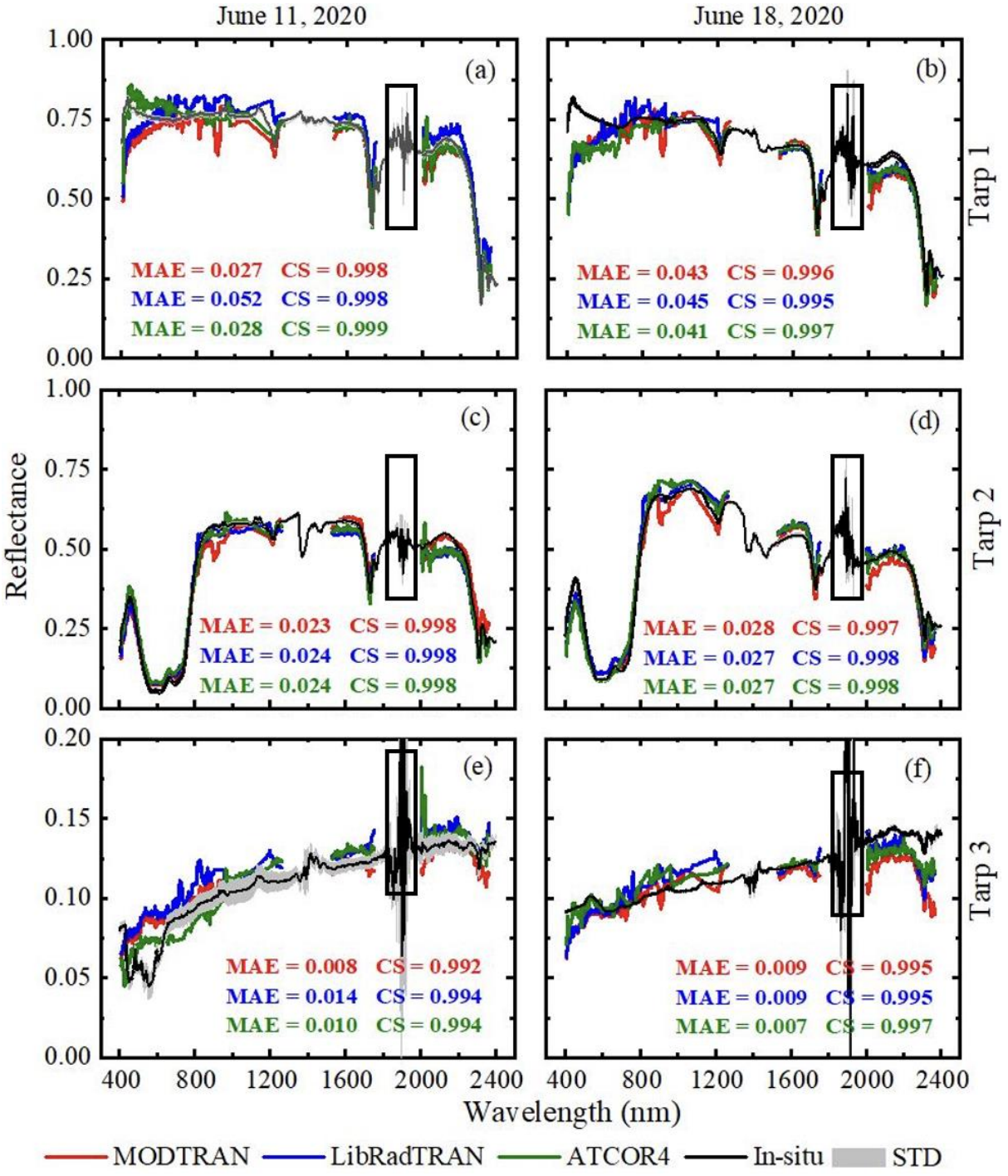


Fig. 3. Reflectance from the Four-Stream-MODTRAN, Four-Stream-LibRadTRAN, ATCOR, and field tarp measurements. The mean absolute error (MAE) and cosine similarity (CS) are also plotted. The

grey areas are standard deviations (STD) of field measurements. Black rectangles indicate bands with large uncertainties.

Surface reflectance from Four-Stream-MODTRAN, Four-Stream-LibRadTRAN, and ATCOR are comparable over soil and vegetation surfaces (Fig. 4). The MAE and CS were calculated based on the results from ATCOR because it is widely used software and can serve as a benchmark. At both land cover types, the results from the three methods are similar with MAE <0.02 and CS >0.99. In the 15×15 m^2 soil and vegetation areas, the uncertainties of surface reflectance from the three methods are comparable (similar STDs). For the soil type, the STD increases from 400 nm to 2400 nm which indicates that soil signals vary more severely in the SWIR region than in the VNIR region. However, STDs are largest at bands in 800–1300 nm, smaller at bands in 1500–2400 nm, and smallest at bands in 400–800 nm for the vegetation land cover type, demonstrating that vegetation signals are more sensitive at 800–1300 nm. Overall, the spectral values and shapes of surface reflectance from the Four-Stream-MODTRAN, Four-Stream-LibRadTRAN, and ATCOR are similar at both soil and vegetation surfaces.

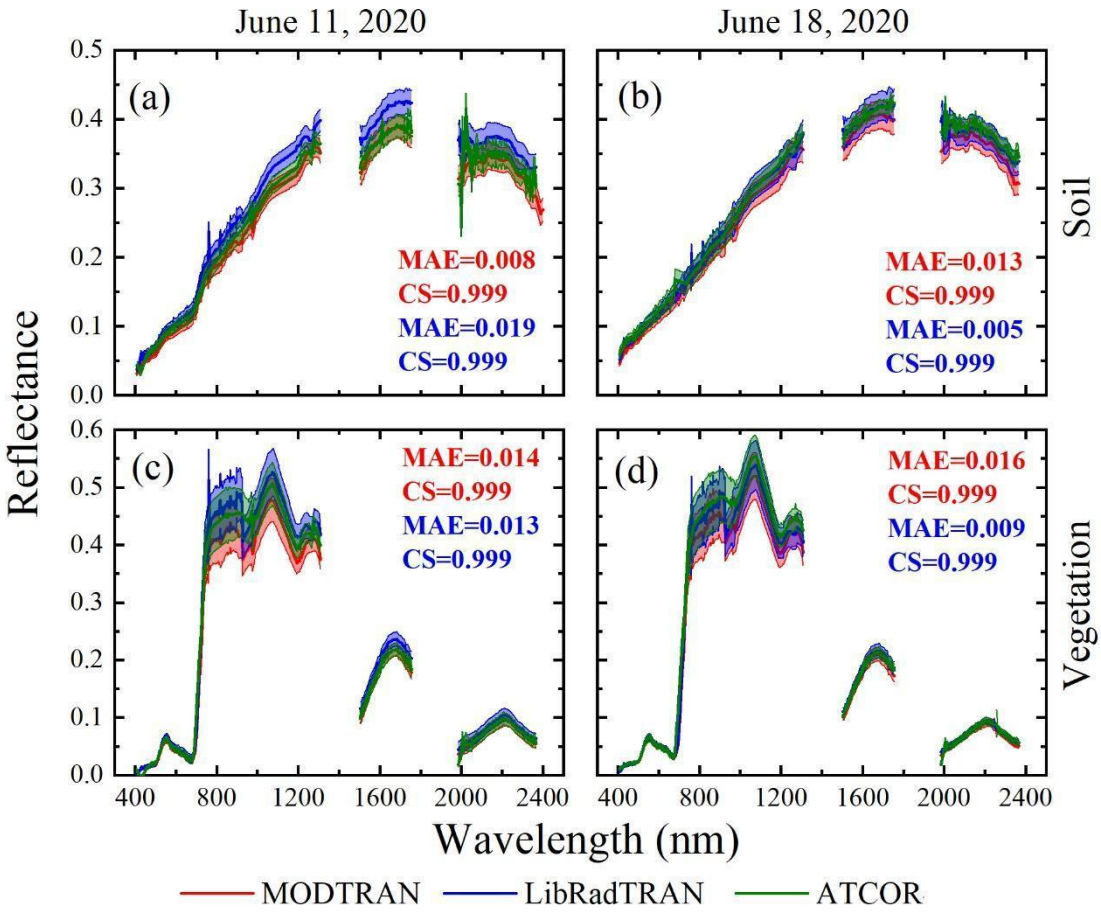


Fig. 4. Reflectance from the Four-Stream-MODTRAN, Four-Stream-LibRadTRAN, and ATCOR of (a) soil and (b) vegetation with a size of 15×15 m^2.

4.2. Accuracies of retrieved atmospheric correction parameters

4.2.1. Sensitivity of atmospheric correction parameters

The parameters' importance for atmospheric correction varies under different conditions. The first-order sensitivity indices (SI) for global sensitivity analysis, which indicate uncertainties of model parameters to the simulated surface reflectance, are shown in Fig. 5. Under comprehensive conditions, the flight altitude and ground elevation have the largest influence on the calculation of surface reflectance. The at-sensor radiance, visibility, water vapor, and relative azimuth angle also have large impacts (Fig. 5(a)). However, if the parameters are measured simultaneously when collecting the AHIS data, these inputs can enable the quantification of uncertainties. With these determined ranges (see Table 4), visibility and water vapor are two dominant factors in producing surface reflectance (Fig. 5(b)). The

sensitivity of visibility is more than 60% for all bands from 400 nm to 2500 nm, and the importance of water vapor can be up to 40% in the atmospheric water absorption windows. The two parameters combined contribute over 95% to sensitivity in calculations of surface reflectance under specific conditions. Thus, accurate retrievals of visibility and water vapor are crucial for the atmospheric correction of AHIS.

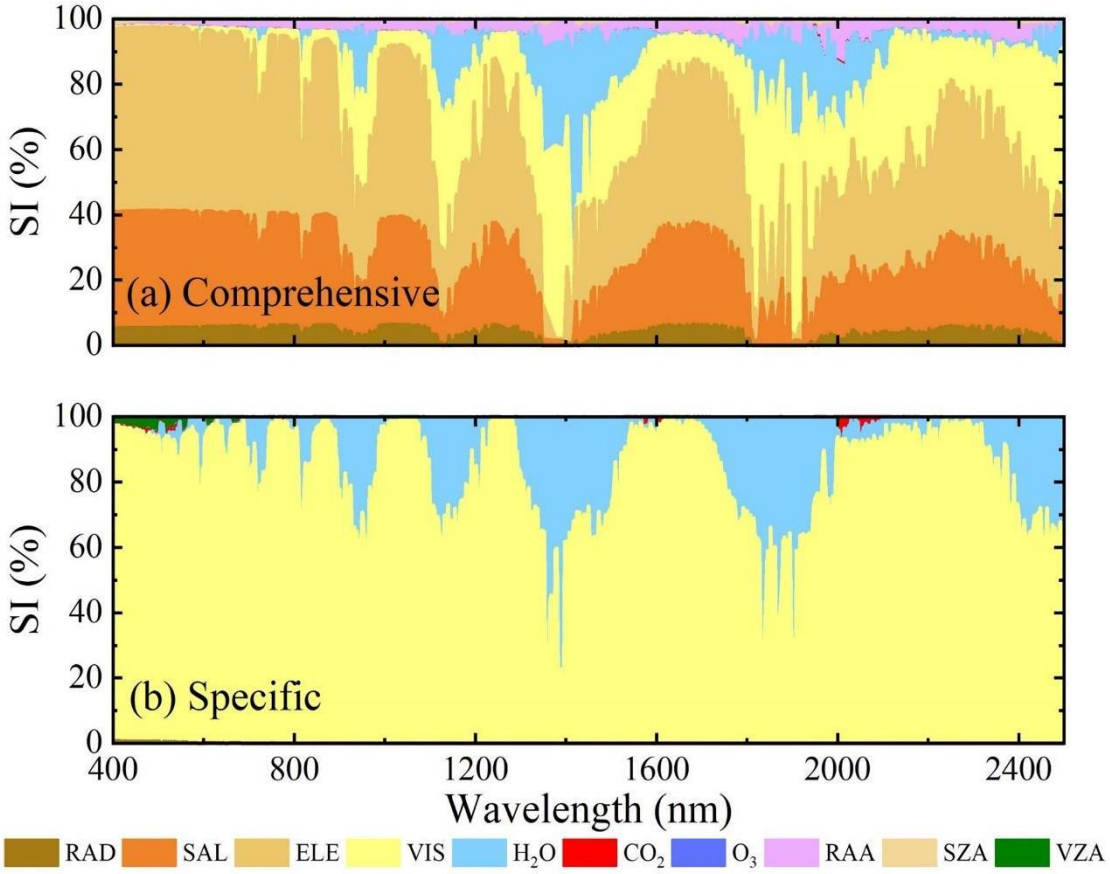


Fig. 5. First-order sensitivity indices (SI) for atmospheric correction parameters of AHIS under (a) comprehensive and (b) specific conditions. The comprehensive and specific conditions refer to unmeasured and measured flight altitude and angles, respectively. RAD: at-sensor radiance; SAL: sensor altitude; ELE: ground elevation; VIS: visibility; H_2O: water vapor; CO_2: carbon dioxide; O_3: ozone; RAA: relative azimuth angle; SZA: Sun zenith angle; VZA: view zenith angle.

Local sensitivity analysis by fixing visibility or water vapor and changing another one indicates that the two parameters influence different bands (Fig. 6). The bands affected by visibility and water vapor at the soil and vegetation land cover types are consistent (columns 1 and 2 in Fig. 6). Specifically,

visibility mainly affects VNIR bands (400–800 nm) and bands near atmospheric windows; water vapor has greater influence at bands near atmospheric windows (1300–1500 nm and 1750–2000 nm). The biases (compared with the mean values) induced by changing visibility and water vapor are larger for vegetation targets than those at the soil surface (columns 3 and 4 in Fig. 6), indicating that vegetation spectra are more sensitive to atmospheric correction and soil spectra are relatively stable. Water vapor causes larger uncertainties than visibility and more attention should be paid to retrieve water vapor for better atmospheric correction of AHIS.

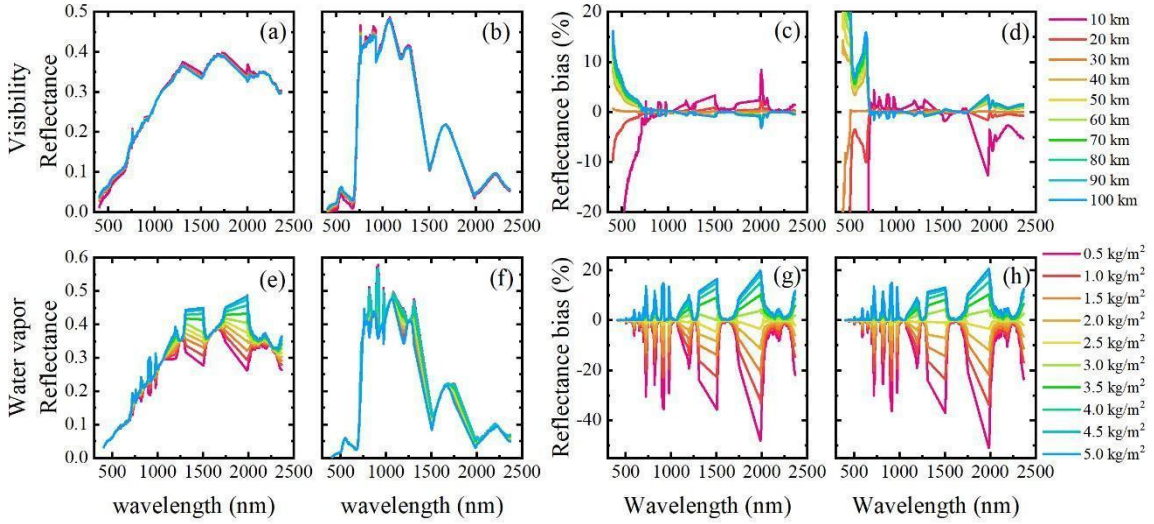


Fig. 6. Reflectance of (a) soil and (b) vegetation with water vapor = 1.8 g/cm^2 and varied visibility, (e) soil and (f) vegetation with visibility = 29.87 km and varied water vapor. The (c) (d), (g) and (h) are reflectance biases of (a), (b), (e), and (f), respectively. Note that the bias is compared with the mean values of the spectra in each subplot.

Visibility and water vapor have large impacts on deriving surface reflectance at different targets and bands. Spatial maps of surface reflectance of two bands (601 nm of VNIR sensor and 2202 nm of SWIR sensor) are displayed for a farm in Fig. 7 to show the influence visually and spatially. As the visibility increases (first and second rows in Fig. 7), the surface reflectance increases at 601 nm (first row) while is nearly unchanged at 2202 nm (second row), indicating that visibility has greater influence at VNIR bands than SWIR bands (similar to Fig. 6). Although the visibility changes at an interval of 10 km for the 601 nm, the increase of surface reflectance is largest from visibility = 10 km to visibility =

20 km. When visibility is larger than 40 km, the increase of surface reflectance with increased visibility is negligible. As the water vapor increases (third and fourth rows in Fig. 7), the surface reflectance increases at 2202 nm (fourth row) and is nearly unchanged at 601 nm (third row). When the water vapor increases from 0.5 g/cm^2 to 4.0 g/cm^2, the surface reflectance at 2202 nm increases; while the water vapor increases from 4.0 g/cm^2 to 5.0 g/cm^2, the increase in surface reflectance is inconspicuous. The visibility influences VNIR bands and water vapor affects SWIR bands, and the variations of visibility >40 km and water vapor >4.0 g/cm^2 generate surface reflectance nearly unchanged. Thus, more attention should be paid to accurately retrieve visibility (<40 km) and water vapor (<4.0 g/cm^2).

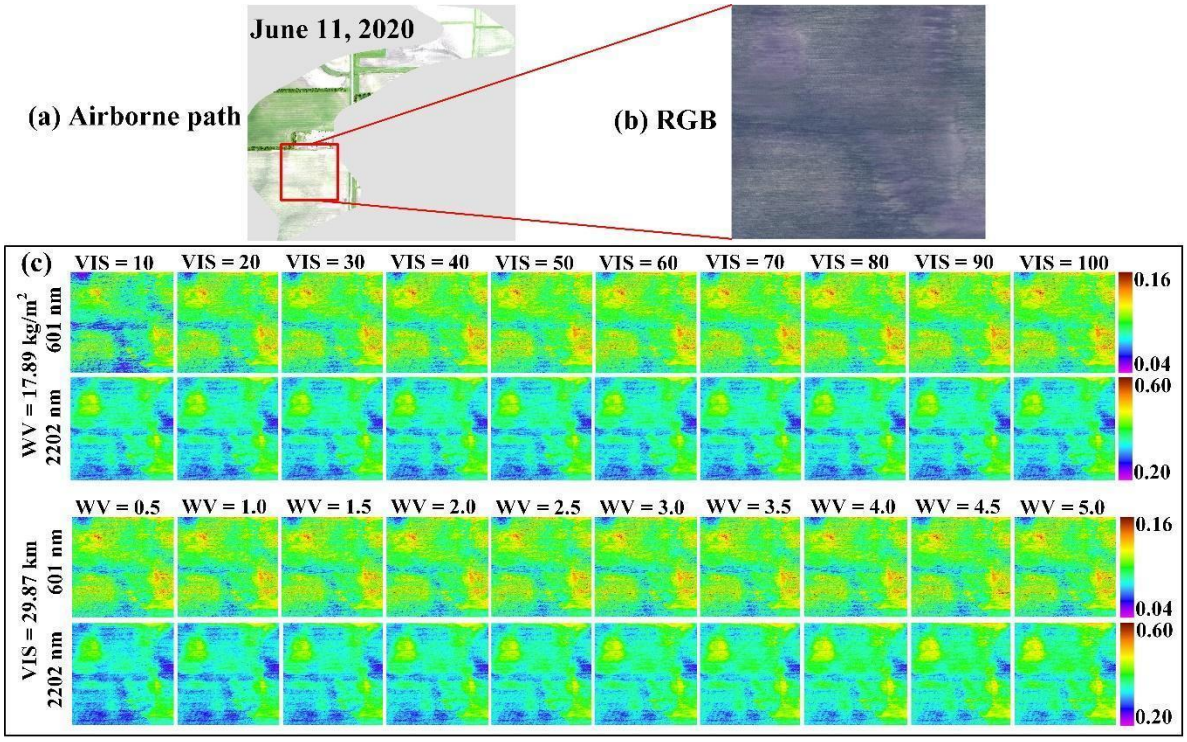


Fig. 7. (a) An example airborne path on June 11th, 2020, (b) selected example farm, and (c) reflectance at 601 nm and 2202 nm of the farm with fixed water vapor (WV, 1.79 g/cm^2) and varied visibility (VIS), and fixed VIS (29.87 km) and varied water vapor.

4.2.2. Performance of atmospheric correction parameters from three sources

The quality of atmospheric parameters from different data sources has a large impact on the accuracy of atmospheric correction. The MAE and CS compared with field tarp measurements for using visibility and water vapor from AHIS, MODIS, and AERONET on June 11th and 18th, 2020 are shown

in Fig. 8. Except for tarp 1 on June 11th, 2020, the visibility and water vapor from AHIS data produce the surface reflectance with the smallest MAE and largest CS. The performances of visibility and water vapor from MODIS (mean R^2 = 0.991; mean RMSE = 0.038) and AERONET (mean R^2 = 0.991; mean RMSE = 0.040) are similar, but worse than those from AHIS data (mean R^2 = 0.994; mean RMSE = 0.028). The results indicate that visibility and water vapor retrieved from AHIS data are the most reliable to perform atmospheric correction of AHIS, although MODIS products and AERONET observations are widely used for atmospheric correction in many studies (Bennouna et al., 2011; Mhawish et al., 2019; Sinyuk et al., 2007).

The visibility and water vapor parameters from the three sources are listed in Table 4. The water vapor is more consistent than visibility among the three datasets. However, large discrepancies exist among the three sources, which may be because of the difference in atmospheric column contents for AHIS (from the surface to flight height), MODIS (from the surface to the top of atmosphere), and AERONET (from the surface to the top of atmosphere). The atmospheric parameters derived from AHIS can better simulate the atmospheric contribution for AHIS images, thus using visibility and water vapor from AHIS directly is recommended for atmospheric correction of AHIS data.

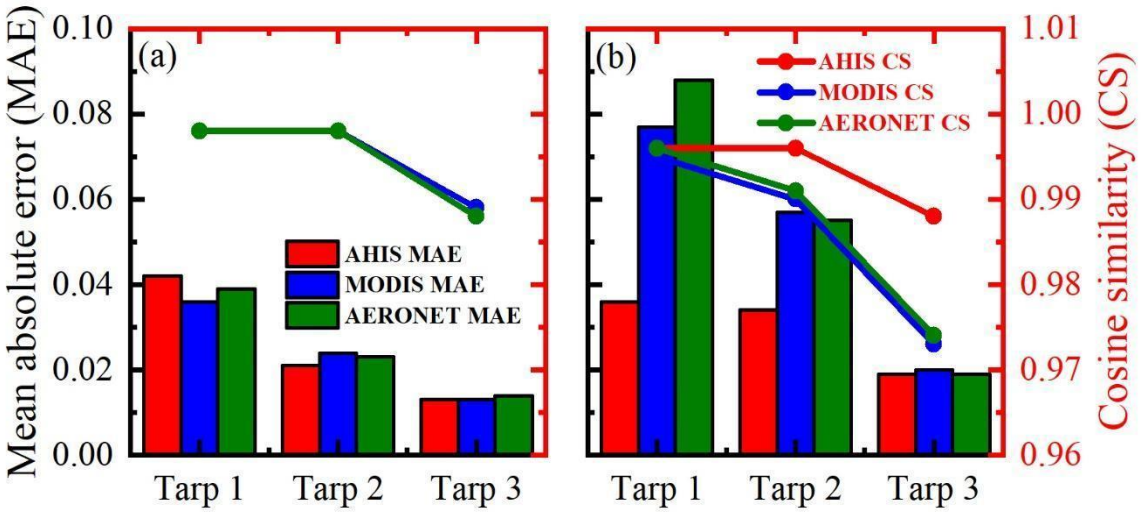


Fig. 8. Mean absolute error (MAE) and cosine similarity (CS) of surface reflectance using visibility and water vapor obtained from AHIS, MODIS, and AERONET on (a) June 11th and (b) June 18th, 2020.

Table 4. Visibility and water vapor derived from AHIS, MODIS products (mean values of nearest nine pixels), and AERONET (Bondville site) observations for atmospheric correction of AHIS data on June 11th and 18th, 2020.

		June 11th, 2020	June 18th, 2020
AHIS	Visibility	29.87 km	40. 48 km
	Water vapor	1.80 g/cm^2	2.27 g/cm^2
MODIS	Visibility	96.04 km	39.65 km
	Water vapor	1.89 g/cm^2	3.43 g/cm^2
AERONET	Visibility	119.95 km	27.78 km
	Water vapor	1.50 g/cm^2	2.27 g/cm^2

4.3. Capability of machine learning for atmospheric correction

Machine learning may have the potential to conduct atmospheric correction of AHIS efficiently and reliably. The performance of Random Forest models to retrieve visibility and water vapor to further derive surface reflectance is shown in Fig. 9. The wavelength importance to retrieve visibility and water vapor is also plotted. Based on the wavelength importance, visibility and water vapor models were built using nine selected bands. The bands used to retrieve visibility were 451 nm, 505 nm, 537 nm, 604 nm, 675 nm, 757 nm, 1264 nm, 2164 nm, and 2300 nm; and the bands for water vapor retrieval were 823 nm, 943 nm, 1131 nm, 1264 nm, 1520 nm, 1615 nm, 2013 nm, 2067 nm, and 2272 nm. The models can retrieve visibility and water vapor accurately with R^2 >0.98, and the retrieval of water vapor was relatively more reliable. Since the variations of visibility and water vapor have a larger influence in deriving surface reflectance (see Fig. 6), the reliable Random Forest models for visibility and water vapor retrievals are useful to generate surface reflectance.

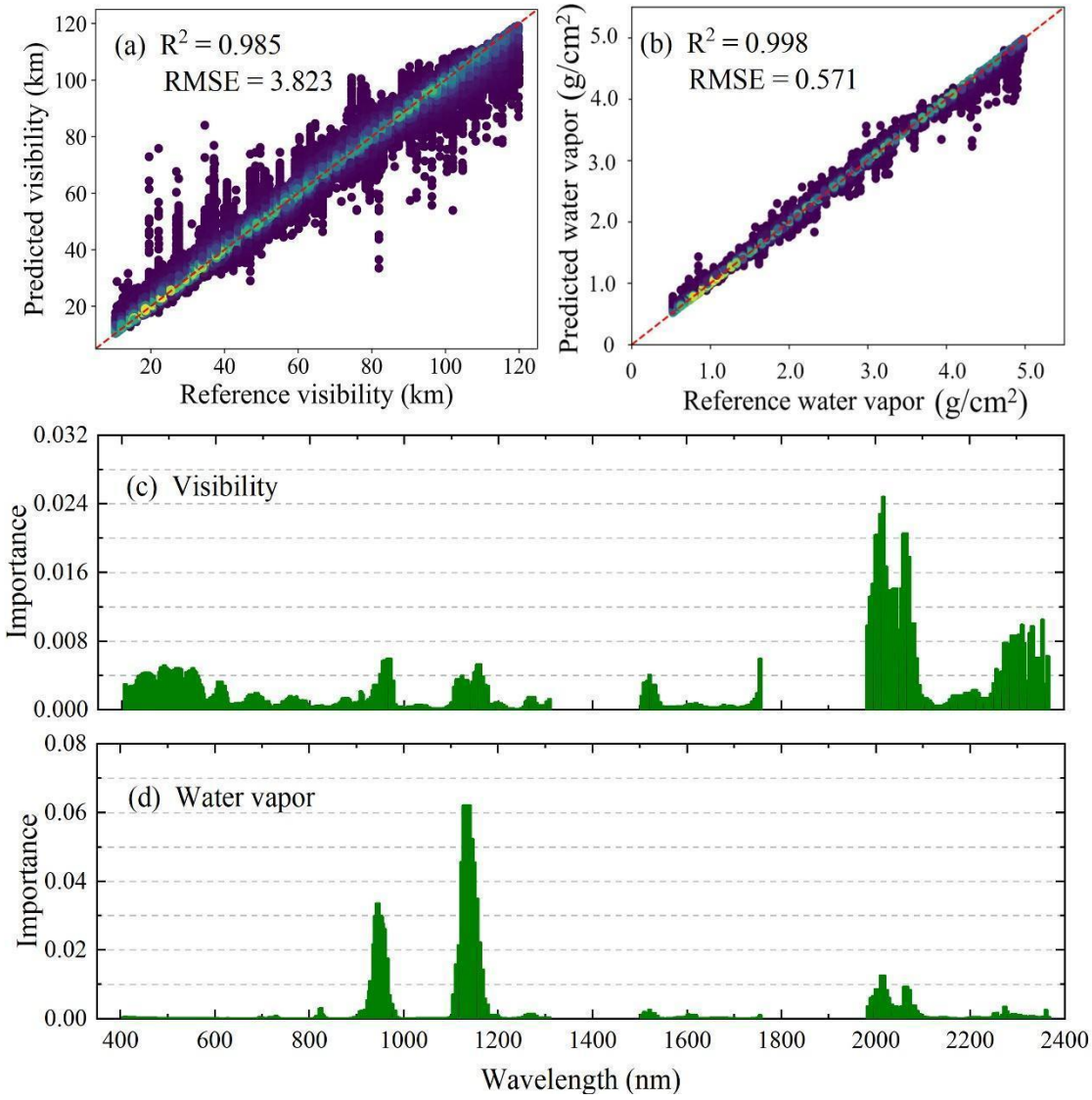


Fig. 9. Performances of Random Forest models for retrieving (a) visibility and (b) water vapor, and the wavelength importance to retrieve (c) visibility and (d) water vapor.

Input parameters in machine learning models are a crucial factor influencing model accuracy. The performance of the Random Forest models for visibility and water vapor retrievals using all AHIS bands, the ATCOR suggested bands (Richter and Schläpfer, 2016), and the selected nine bands based on wavelength importance are listed in Table 5. Using all bands as the input obtained the best performance for retrieving visibility while the worst results for water vapor. Because water vapor is sensitive to several key bands, redundant information, as well as irrelevant bands, likely reduce model performance. The suggested bands in ATCOR are based on empirical relationships (Richter and Schläpfer, 2016),

which could get reasonable results. However, if the bands were selected based on wavelength importance rather than empirical knowledge, the model performance can be improved. To reduce inputs of machine learning models and improve their accuracy, we suggest using wavelength importance as an indicator to select suitable inputs for model developments. Notably, the bands with the highest importance for water vapor in the Random Forest model (943 and 1131 nm) correspond to atmospheric absorption features that are used for water vapor retrieval in ATCOR (which also uses a feature around 823 nm, also identified in our Random Forest model), which indicates that empirical bands can obtain relatively good estimations and machine learning can improve.

Table 5. Performance of Random Forest models to retrieve visibility and water vapor using three different band combinations including all bands, bands suggested by the ATCOR, and bands selected

Variable	**Metrics**	**All bands[1]**	**ATCOR[2]**	**Selected bands[3]**
Visibility	R^2	0.988	0.984	0.985
	RMSE (km)	3.508	3.964	3.823
Water vapor	R^2	0.995	0.997	0.998
	RMSE (kg/m^2)	0.865	0.716	0.571

[1]All bands of AHIS are used for visibility and water vapor models;

[2]Visibility model: 480 nm, 550 nm, 660 nm, 850 nm, 1600 nm, and 2200 nm; water vapor model: 800 nm, 820 nm, 840 nm, 890 nm, 940 nm, 1010 nm, 1080 nm, 1130 nm, and 1220 nm.

[3]Visibility model: 451 nm, 505 nm, 537 nm, 604 nm, 675 nm, 757 nm, 1264 nm, 2164 nm, and 2300 nm; and water vapor model: 823 nm, 943 nm, 1131 nm, 1264 nm, 1520 nm, 1615 nm, 2013 nm, 2067 nm, and 2272 nm.

The performance of Random Forest-based atmospheric correction is shown in Fig. 10 with ten-fold cross-validation, and the R^2 and RMSE at each AHIS band are plotted in Fig. 10(a). The R^2 was >0.99 for most bands and >0.98 for all bands, and the RMSE was <0.03 for most bands and <0.05 for all bands. The performance of Random Forest-based atmospheric correction was highly related to the variations of at-sensor radiance simulated by MODTRAN (orange points in Fig. 10(a)). Due to the influence of

gaseous absorption, the radiance is affected severely at some bands (e.g. around 940–1120 nm), and the Random Forest-based atmospheric correction performed relatively worse at these bands. However, the R^2 was still >0.98 and RMSE <0.05 in these wavelengths, such that the worst performance of Random Forest-based atmospheric correction at gaseous absorption bands was also acceptable.

Four bands (531 nm, 1144 nm, 1630 nm, and 2202 nm) were selected to analyze the results of Random Forest-based atmospheric correction for normal bands in three spectral regions (531 nm, 1630 nm, and 2202 nm) and affected bands (1144 nm) in gaseous absorption regions. For the less affected bands from VNIR to SWIR bands (Fig. 10(a), (d), and (e)), the Random Forest prediction and reference points were more evenly distributed along the 1:1 lines (red dash lines in Fig. 10), and the R^2 was >0.99 and RMSE <0.02. However, for the gaseous absorption band (Fig. 10(c)), the performance was slightly worse, and the points were more scattered (e.g. the red circles in Fig. 10(c)). The R^2 decreased to 0.98 and RMSE increased to 0.05, which is more than double those at unaffected bands, indicating that machine learning performs better for atmospheric correction of AHIS at non-absorption bands. Although large dynamic variations occur at gaseous absorption bands due to the influence of atmospheric gases, the performance was still encouraging with a high R^2 (>0.98) and a low RMSE (<0.05). Machine learning shows great potential to conduct atmospheric correction for all bands of AHIS after removing water vapor channels.

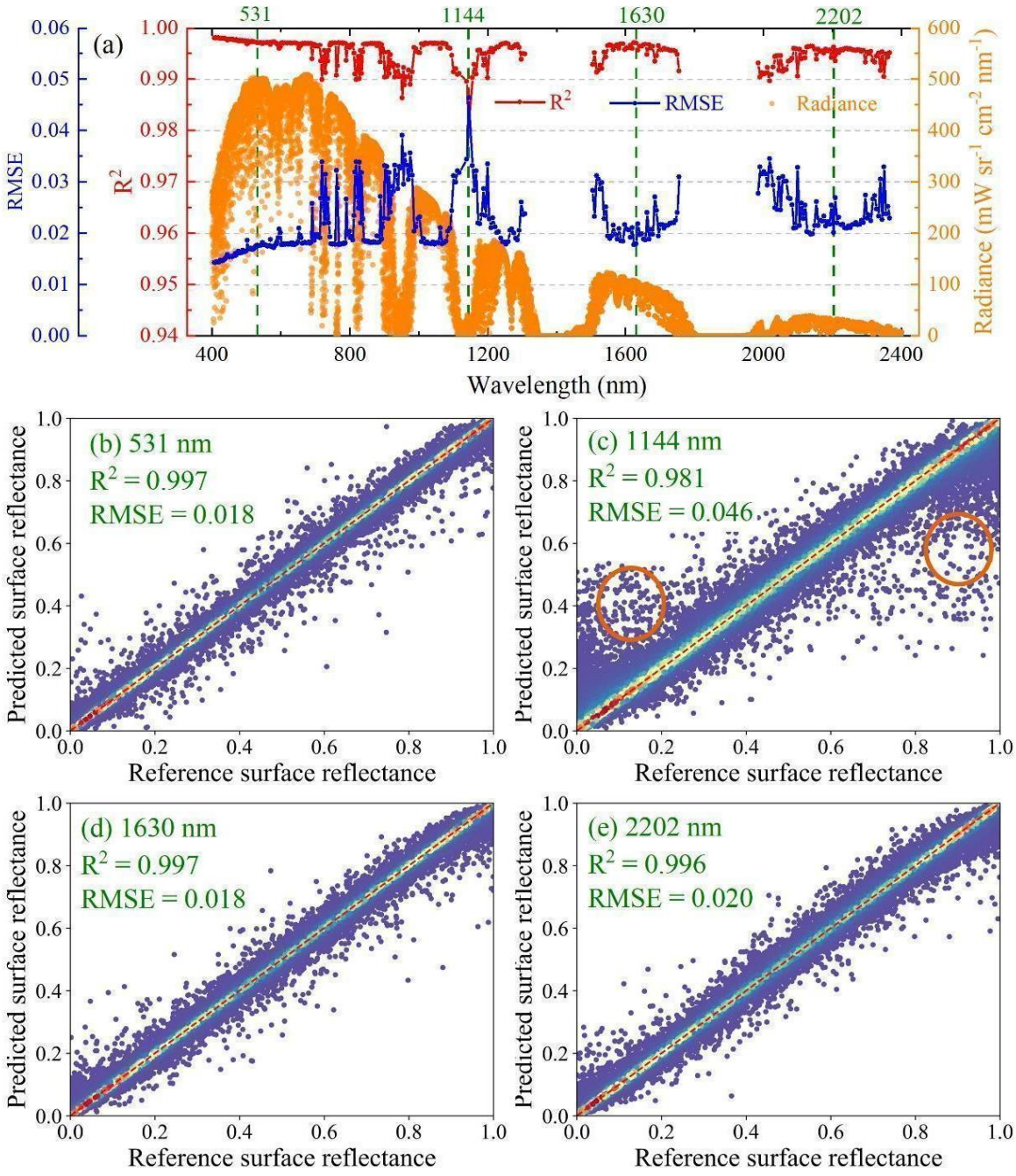


Fig. 10. Ten-fold cross-validation of Random Forest-based models for atmospheric correction of AHIS using MODTRAN simulated datasets: (a) of all bands, (b) at 531 nm, and (c) at 1144 nm, (d) at 1630 nm, and (e) 2202 nm. The red dash lines are the 1:1 lines and red circles indicate predictions with large biases.

After a combination of submodel #1 and submodel #2, surface reflectance can be derived from AHIS at-sensor radiance directly. The comparison of machine learning-derived and traditional RTM-

derived hyperspectral surface reflectance is shown in Fig.11. For the three tarps (from high to low reflective properties), the two kinds of surface reflectance agree well (align closely to the 1:1 line in Figs.11 a and c) with MAE<0.03 and CS>0.99 (Figs.11 b and d). Generally, the performance in the middle spectra (700 nm to 1300 nm) is a little bit poorer than the other wavelengths, which is consistent with the performance in the machine learning model development (shown in Fig. 10a). The results indicate that the machine learning-based atmospheric correction using the relatively simple Random Forest model can achieve comparable accuracy to RTM-based models. Since the machine learning models were developed based on RTM simulations, the performance compared with field tarp measurements was slightly decreased (Fig. S4). The machine learning model was designed to speed up the data processing and it was acceptable to have a similar performance as RTM-based atmospheric correction. More complex machine learning or deep learning models such as deep neural networks, recurrent neural networks, and convolutional neural networks may have the potential to achieve better performance (Zhu et al., 2017).

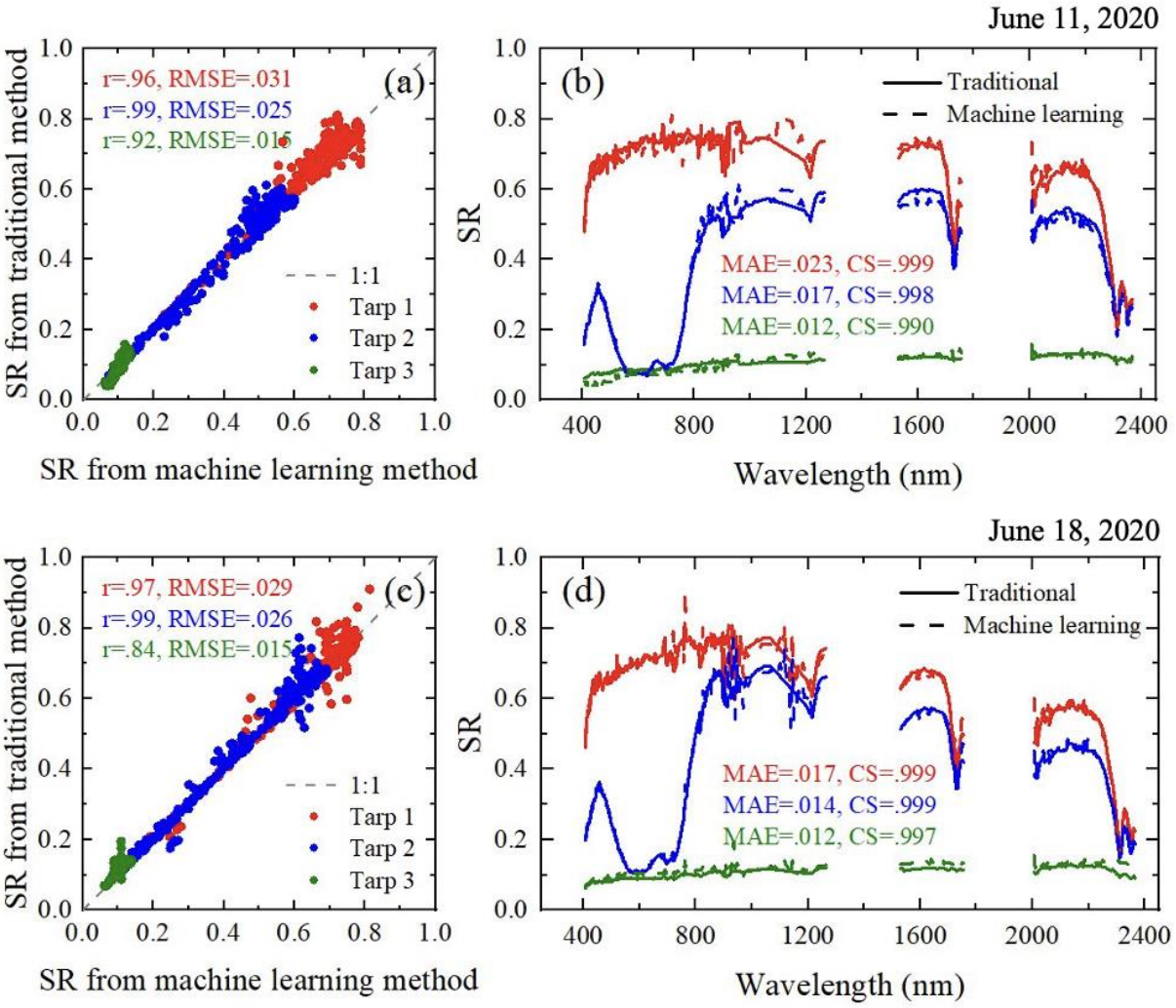


Fig. 11. Validation of surface reflectance from machine learning-based and traditional RTM-based atmospheric correction on June 11th and 18th, 2020 over tarp 1, tarp 2, and tarp 3.

5. Discussion

This study proposed an automated and operational pipeline for atmospheric correction of AHIS. The accuracy was enhanced by selecting suitable atmospheric correction solutions including the Four-Stream-MODTRAN and Four-Stream-LibRadTRAN benchmarked with ATCOR, identifying the importance of atmospheric correction parameters, and comparing the performance of parameters retrieved directly from AHIS, or alternatively from MODIS or AERONET. The processing efficiency was improved by applying machine learning emulators. The performance comparison of the developed pipeline with other studies, along with its opportunities and limitations, are discussed below.

5.1. Four-stream radiative transfer theory for airborne remote sensing

Four-stream radiative theory in previous studies is designed for satellite sensors (Verhoef, 1985; Verhoef et al., 2018; Verhoef and Bach, 2012, 2007, 2003). The potential of the four-stream radiative theory for airborne sensors is rarely explored. Atmospheric correction for an airborne platform is more complicated than for sensors from space because the sensor is located inside the atmosphere with signals from one part above and another part below it. By modification of the four-stream radiative transfer theory for airborne sensors, atmospheric correction for airborne observations can be conducted using the four-stream theory (see Appendix for details). Theoretically, adjacent effects should be considered for high-resolution observations since the scene is small (Fawcett et al., 2018). However, the calculation of background reflectance is relatively complex, and using smoothed reflectance values (e.g. scene average) is common practice (Fawcett et al., 2018). The results of the four-stream theory assuming a uniform or non-uniform surface are similar for the agricultural region which is relatively flat and homogeneous (see Fig. S3). Thus, the four-stream theory is feasible to atmospherically correct AHIS data for the agricultural fields.

The developed pipeline for atmospheric correction can achieve high accuracy in producing surface reflectance with mean absolute errors of <0.05 and cosine similarities >0.98. Such performance outperformed the reported accuracies in studies such as Dao et al., (2019), Katkovsky et al., (2018), and Rani et al., (2017). Besides the model performance, the pipeline can be flexibly coupled with surface radiative transfer models to directly quantify surface attributes from at-sensor radiance (Verhoef et al., 2018; Verhoef and Bach, 2012, 2007, 2003). Also, solar-induced fluorescence and thermal infrared emissions can be included in the pipeline because the four-stream theory can consider these signals to solve radiative transfer problems in multiple systems such as soil-water, soil-vegetation, soil-water-atmosphere, and soil-vegetation-atmosphere systems (Verhoef et al., 2018, 2007; Yang et al., 2020a).

5.2. Future improvements and applications

The performance of the proposed pipeline is accurate and promising, and several aspects could be improved to broaden its applications. First, the pipeline is designed for stable atmospheric illumination

conditions and agricultural ecosystems, which might be limited for applications in regions with frequent and variable cloudy conditions such as tropics (Zhou et al., 2020a) and high latitudes (Wang et al., 2018). The uncertainty introduced by clouds (flight height above the clouds) or cloud shadows (flight height below the clouds) needs to be further corrected through novel algorithms (Wang et al., 2019). Second, BRDF and adjacent effects should be considered in the pipeline for the applications of UAS and satellite data. The view zenith angle of satellite images could be up to 40° even to 70° (Zhou et al., 2020b), while it is generally less than 17° for the AHIS system. BRDF at larger view zenith angles and adjacent effects at less homogenous regions is more severe, which cannot be ignored for atmospheric correction. Third, the Random Forest algorithm was used in machine learning emulators for atmospheric correction. More advanced machine learning methods, such as deep neural networks, recurrent neural networks, and convolutional neural networks, should be compared to achieve better performance (Zhu et al., 2017). Finally, some modifications should be included when the pipeline is applied for large regions or regions without "dark" pixels. The visibility (or aerosol optical depth) in this pipeline is estimated based on DDV pixels, which should be modified for regions without "dark" pixels. The average value of visibility and water vapor retrievals (image-wise processing) are used for stability and efficiency in this study because the airborne scene only covers a small area. When an airborne survey covers larger or less-homogenous regions, spatial variations of these parameters (pixel-wise processing) should be considered.

Besides manned airplane systems, this pipeline to atmospherically correct AHIS data can also be applied to process hyperspectral data from other platforms such as unmanned aerial systems (UAS) and satellites. Particularly for the forthcoming satellite hyperspectral missions, such as NASA's SBG (Surface, Biology, and Geology) (Thompson et al., 2020), PACE (Plankton, Aerosol, Cloud, and ocean Ecosystem) (Hasekamp et al., 2019), and ESA's CHIME (Copernicus Hyperspectral Imaging Mission for the Environment) (Nieke and Rast, 2018).

6. Conclusions

This study developed an automated and operational processing pipeline to derive surface reflectance from large-volume AHIS imagery. Field tarp hyperspectral measurements for targets with low, medium, and high reflective properties were used for comprehensive validations of surface reflectance derived from the pipeline. Based on the four-stream radiative transfer theory, the pipeline could generate surface reflectance accurately (MAE <0.05 and CS >0.98) with low processing overhead, which is comparable to the results using ATCOR software. Sensitivity analysis found visibility and water vapor are two key parameters in the atmospheric correction which should be derived from AHIS data directly rather than from MODIS or AERONET observations. We suggested a suite of spectral wavelengths to retrieve atmospheric visibility and water vapor using wavelength importance as an indicator. Random Forest models were built to produce accurate visibility and water vapor retrievals and surface reflectance, which do not involve RTMs and precalculated LUTs, and have the potential to speed up the atmospheric correction of AHIS. Atmospheric correction based on the four-stream theory shown in the Appendix is reliable and accurate for atmospheric correction of sensors inside and outside the atmosphere. The universal atmospheric correction embedded in the developed pipeline has great potential for processing hyperspectral imagery from various platforms such as UAS and satellites.

Description of author's responsibilities:

Q.Z., S.W., and K.G. conceived the project. K.G., S.W., and P.T. contributed to funding acquisition. Q.Z. and S.W. performed the experiments and data collection. Q.Z., S.W., and N.L. conducted data processing and analysis. Q.Z., S.W., N.L., P.T., C.J., B.P., W.V., and K.G. contributed to data interpretation and manuscript discussion. W.V. developed the four-stream-based atmospheric correction for airborne hyperspectral data in the appendix, which laid a solid foundation for atmospheric correction in this work. Q.Z. wrote the original draft of the manuscript. All authors have revised and approved the final manuscript.

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Appendix. Atmospheric correction of AHIS using the four-stream theory

In the four-stream system, radiative fluxes from the land surface and in the atmosphere can be divided into four components: direct solar flux, diffuse downward flux, diffuse upward flux, and directly observed flux (radiance). Scattering by a layer and reflectance by a surface can be described with vectors and matrices (Verhoef, 1985):

$$\begin{pmatrix} E_s(\mathrm{b}) \\ E^-(\mathrm{b}) \\ E^+(\mathrm{t}) \\ E_o(\mathrm{t}) \end{pmatrix} = \begin{pmatrix} \tau_{ss} & 0 & 0 & 0 \\ \tau_{sd} & \tau_{dd} & \rho_{dd} & 0 \\ \rho_{sd} & \rho_{dd} & \tau_{dd} & 0 \\ \rho_{so} & \rho_{do} & \tau_{do} & \tau_{oo} \end{pmatrix} \begin{pmatrix} E_s(\mathrm{t}) \\ E^-(\mathrm{t}) \\ E^+(\mathrm{b}) \\ E_o(\mathrm{b}) \end{pmatrix} \text{ or } \begin{pmatrix} \mathbf{E}^-(\mathrm{b}) \\ \mathbf{E}^+(\mathrm{t}) \end{pmatrix} = \begin{pmatrix} \mathbf{T_d} & \mathbf{R_b} \\ \mathbf{R_t} & \mathbf{T_u} \end{pmatrix} \begin{pmatrix} \mathbf{E}^-(\mathrm{t}) \\ \mathbf{E}^+(\mathrm{b}) \end{pmatrix} \quad \text{(A1)}$$

where (b) and (t) indicate the bottom and the top of the layer, respectively. The layer scattering matrix includes a downward transmittance matrix, an upward transmittance matrix, a reflectance at the top, and one at the bottom. For a non-Lambertian surface, we can write $\begin{pmatrix} E^+ \\ E_o \end{pmatrix} = \begin{pmatrix} r_{sd} & r_{dd} \\ r_{so} & r_{do} \end{pmatrix} \begin{pmatrix} E_s \\ E^- \end{pmatrix}$, or in matrix-vector notation $\mathbf{E}^+ = \mathbf{R}\mathbf{E}^-$. A layer can be combined with a surface at the bottom as follows:

$$\begin{aligned} \mathbf{E}^-(\mathrm{b}) &= \mathbf{T_d}\mathbf{E}^-(\mathrm{t}) + \mathbf{R_b}\mathbf{E}^-(\mathrm{b}) \\ \mathbf{E}^+(\mathrm{b}) &= \mathbf{R}\mathbf{E}^-(\mathrm{b}) \end{aligned} \quad \text{(A2)}$$

Thus, we can derive the matrix of the combination layer-surface:

$$\mathbf{E}^+(\mathrm{t}) = \mathbf{R_t}\mathbf{E}^-(\mathrm{t}) + \mathbf{T_u}\mathbf{E}^+(\mathrm{b}) = [\mathbf{R_t} + \mathbf{R}(\mathbf{I} - \mathbf{R_b}\mathbf{R})^{-1}\mathbf{T_d}]\mathbf{E}^-(\mathrm{t}) \quad \text{(A3)}$$

To derive the radiance sensed on board an airborne platform flying at some altitude in the atmosphere, for a uniform Lambertian surface with albedo *a*, the same four-stream system can also be applied if two atmospheric layers are considered, one above, and one below the platform. The level of the earth's surface, airborne platform, and TOA are called 2, 1, and 0, respectively The atmospheric layer above the airborne platform is labeled 1, the one under airborne platform 2, and for the whole atmosphere, no label is used. Thus, we can write $\mathbf{E}^-(2) = \mathbf{T_d}\mathbf{E}^-(0) + \mathbf{R_b}\mathbf{E}^+(2)$, and since $\mathbf{E}^+(2) = \mathbf{A}\mathbf{E}^-(2)$, pre-multiplication of the first equation by $\mathbf{A}$ gives $\mathbf{A}\mathbf{E}^-(2) = \mathbf{E}^+(2) = \mathbf{A}\mathbf{T_d}\mathbf{E}^-(0) + \mathbf{A}\mathbf{R_b}\mathbf{E}^+(2)$, thus:

$$\mathbf{E}^+(2) = (\mathbf{I} - \mathbf{A}\mathbf{R_b})^{-1}\mathbf{A}\mathbf{T_d}\mathbf{E}^-(0) \tag{A4}$$

For the level of the airborne platform, we apply the following equations:

$$\begin{matrix}\mathbf{E}^-(1)=\mathbf{T_{d1}}\mathbf{E}^-(0)+\mathbf{R_{b1}}\mathbf{E}^+(1)\\ \mathbf{E}^+(1)=\mathbf{R_{t2}}\mathbf{E}^-(1)+\mathbf{T_{u2}}\mathbf{E}^+(2)\end{matrix} \tag{A5}$$

Applying the first equation of A5 and A4 into the second equation of A5, we obtain:

$$\mathbf{E}^+(1) = (\mathbf{I} - \mathbf{R_{t2}}\mathbf{R_{b1}})^{-1}[\mathbf{R_{t2}}\mathbf{T_{d1}} + \mathbf{T_{u2}}(\mathbf{I} - \mathbf{A}\mathbf{R_b})^{-1}\mathbf{A}\mathbf{T_d}]\mathbf{E}^-(0) \tag{A6}$$

For the elaboration of A6, we first simplify two expressions by eliminating the matrix inverses:

$$\begin{aligned}(\mathbf{I} - \mathbf{A}\mathbf{R_b})^{-1}\mathbf{A} &= \left[\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} - \begin{pmatrix}a & a\\ a & a\end{pmatrix}\begin{pmatrix}0 & 0\\ \rho_{dd} & 0\end{pmatrix}\right]^{-1}\begin{pmatrix}a & a\\ a & a\end{pmatrix}\\ &= \frac{a}{1-a\rho_{dd}}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}\end{aligned} \tag{A7}$$

$$\begin{aligned}(\mathbf{I} - \mathbf{R_{t2}}\mathbf{R_{b1}})^{-1} &= \left[\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} - \begin{pmatrix}\rho_{sd2} & \rho_{dd2}\\ \rho_{so2} & \rho_{do2}\end{pmatrix}\begin{pmatrix}0 & 0\\ \rho_{dd1} & 0\end{pmatrix}\right]^{-1}\\ &= \frac{1}{1-\rho_{dd2}\rho_{dd1}}\begin{pmatrix}1 & 0\\ \rho_{do2}\rho_{dd1} & 1-\rho_{dd2}\rho_{dd1}\end{pmatrix}\end{aligned} \tag{A8}$$

Thus, Eq. (A6) can be written as:

$$\begin{aligned}\mathbf{E}^+(1) &= \frac{E_s(0)\begin{pmatrix}1 & 0\\ \rho_{do2}\rho_{dd1} & 1-\rho_{dd2}\rho_{dd1}\end{pmatrix}}{1-\rho_{dd2}\rho_{dd1}}\left[\begin{pmatrix}\rho_{sd2} & \rho_{dd2}\\ \rho_{so2} & \rho_{do2}\end{pmatrix}\begin{pmatrix}\tau_{ss1} & 0\\ \tau_{sd1} & \tau_{dd1}\end{pmatrix} + \begin{pmatrix}\tau_{dd2} & 0\\ \tau_{do2} & \tau_{oo2}\end{pmatrix}\begin{pmatrix}1\\ 1\end{pmatrix}\frac{a(\tau_{ss}+\tau_{sd})}{1-a\rho_{dd}}\right]\\ &= \begin{pmatrix}\frac{\rho_{sd2}\tau_{ss1}+\rho_{dd2}\tau_{sd1}+\tau_{dd2}a(\tau_{ss}+\tau_{sd})/(1-a\rho_{dd})}{1-\rho_{dd2}\rho_{dd1}}E_s(0)\\ E_0(1)\end{pmatrix}\end{aligned} \tag{A9}$$

where $E_0(1) = \pi L_o(1)$, and $L_o(1)$ is the sensed radiance at the level of the airborne platform. By a complete elaboration, the following expression for $L_o(1)$ is obtained:

$$L_o(1) = [\tau_{ss1}\rho_{so2} + \frac{(\tau_{sd1}+\tau_{ss1}\rho_{sd2}\rho_{dd1})\rho_{do2}}{1-\rho_{dd2}\rho_{dd1}} + \frac{(\tau_{sd}+\tau_{ss})(\tau_{oo2}+\tau_{do2}+\frac{\tau_{dd2}\rho_{dd1}\rho_{do2}}{1-\rho_{dd2}\rho_{dd1}})a}{1-a\rho_{dd}}]\frac{E_s^o\cos\theta_s}{\pi} \tag{A10}$$

This result looks very complex, but considered as a function of the surface albedo a, it can be reduced to an equation that is identical to the one for TOA observations, and Eq. (A10) therefore can be simplified into:

$$L_{\mathrm{o}}(1) = L_{\mathrm{p0}}(1) + \frac{G_t+G_b}{1-aS}a \tag{A11}$$

where G_t and G_b are the gain factors for target and background, respectively, S is the spherical albedo of the complete atmosphere, and $L_{\mathrm{p0}}(1)$ is the path radiance at sensor height. (Verhoef and Bach, 2012, 2007, 2003) developed the MODTRAN Interrogation Technique, which runs MODTRAN for a Lambertian surface with surface albedos (a) of 0%, 50%, and 100%, respectively.

For a non-uniform Lambertian surface, replacing $\mathbf{A} = \begin{pmatrix} a & a \\ a & a \end{pmatrix}$ with $\mathbf{R} = \begin{pmatrix} r_b & r_b \\ r_t & r_t \end{pmatrix}$, with target reflectance $r_t = r_{so} = r_{do}$ and background adjacent reflectance $r_b = \overline{r_{sd}} = \overline{r_{dd}}$ (Fawcett et al., 2018), Eq. (A10) should be modified into:

$$L_{\mathrm{o}}(1) = \left[\tau_{ss1}\rho_{so2} + \frac{(\tau_{sd1}+\tau_{ss1}\rho_{sd2}\rho_{dd1})\rho_{do2}}{1-\rho_{dd2}\rho_{dd1}} + \frac{(\tau_{sd}+\tau_{ss})[r_t\tau_{oo2}+r_b(\tau_{do2}+\frac{\tau_{dd2}\rho_{dd1}\rho_{do2}}{1-\rho_{dd2}\rho_{dd1}})]}{1-r_b\rho_{dd}}\right]\frac{E_S^o\cos\theta_s}{\pi} \tag{A12}$$

which can be reduced to:

$$L_{\mathrm{o}}(1) = L_{\mathrm{p0}}(1) + \frac{G_t r_t+G_b r_b}{1-Sr_b} \tag{A13}$$

If the background reflectance is taken to be equal to the scene average, then the scene average radiance becomes:

$$\overline{L_{\mathrm{o}}(1)} = L_{\mathrm{p0}}(1) + \frac{G_t+G_b}{1-Sr_b}r_b \tag{A14}$$

The background reflectance (r_b) is then found as $\frac{\overline{L_{\mathrm{o}}(1)}-L_{\mathrm{p0}}(1)}{G_t+G_b+S[\overline{L_{\mathrm{o}}(1)}-L_{\mathrm{p0}}(1)]}$, and finally, the atmospheric correction of a non-uniform Lambertian surface is accomplished by:

$$r_t = \frac{[L_{\mathrm{o}}(1)-L_{\mathrm{p0}}(1)](1-Sr_b)-G_b r_b}{G_t} \tag{A15}$$

The MODTRAN outputs provide: (1) PATH (radiation scattered by the atmosphere and including adjacency effects, $L_{\mathrm{p0}}(1) + \frac{G_b a}{1-\mathrm{a}S}$) and (2) GRND (ground reflected radiation directly transmitted to the sensor, $\frac{G_t a}{1-\mathrm{a}S}$). In this particular case, two runs with MODTRAN for surface albedos of 0.5 and 1 are sufficient to derive the constants S, G_t, G_b and $L_{\mathrm{p0}}(1)$. Using the percentages albedo as subscripts, we

can obtain: $\frac{G_t}{2-S} = \mathrm{GRND}_{50}$, $\frac{G_t}{1-S} = \mathrm{GRND}_{100}$, $L_{\mathrm{p0}}(1) + \frac{G_b}{2-S} = \mathrm{PATH}_{50}$, and $L_{\mathrm{p0}}(1) + \frac{G_b}{1-S} = \mathrm{PATH}_{100}$. From these, we can obtain the parameters for atmospheric correction for a non-uniform Lambertian surface as:

$$S = \frac{\mathrm{GRND}_{100} - 2\mathrm{GRND}_{50}}{\mathrm{GRND}_{100} - \mathrm{GRND}_{50}} \tag{A16}$$

$$G_t = \frac{\mathrm{GRND}_{100}\mathrm{GRND}_{50}}{\mathrm{GRND}_{100} - \mathrm{GRND}_{50}} \tag{A17}$$

$$G_b = \frac{\mathrm{PATH}_{100} - \mathrm{PATH}_{50}}{\mathrm{GRND}_{100} - \mathrm{GRND}_{50}} G_t \tag{A18}$$

$$L_{\mathrm{p0}}(1) = \frac{\mathrm{PATH}_{50}\mathrm{GRND}_{100} - \mathrm{PATH}_{100}\mathrm{GRND}_{50}}{\mathrm{GRND}_{100} - \mathrm{GRND}_{50}} \tag{A19}$$